

Question 1

Not yet answered

Marked out of 1.00

k-NN can be used for both classification and regression.

Select one:

- ☐ True
- ☐ False

Question 2

Not yet answered

Marked out of 1.00

An association rule $X \rightarrow Y$ does not mean that X causes Y.

- ☐ True
- ☐ False

Question 3

Not yet answered

Marked out of 2.00

What are the limitations of the min-max normalization method?

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A ▾

B

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x_2

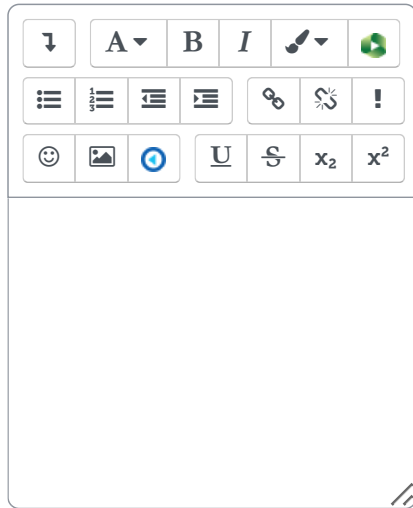
x^2

Question 4

Not yet answered

Marked out of 3.00

Qualitatively describe what mode collapse is in a generative adversarial network (GAN).

A rich text editor interface with a toolbar at the top containing icons for bold, italic, underline, link, unlink, and list. Below the toolbar is a large, empty text area for the user's response.**Question 5**

Not yet answered

Marked out of 1.00

What is a key challenge addressed by EXT5 in multi-task learning?

Select one:

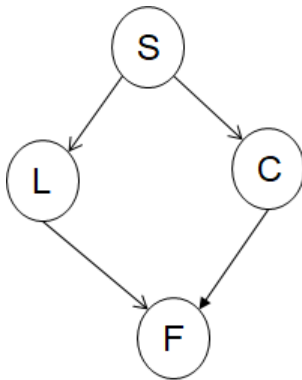
- ☐ a. Overfitting to unseen tasks
- ☐ b. Eliminating unsupervised learning
- ☐ c. Negative transfer between tasks
- ☐ d. Reducing the size of the model

Question 6

Not yet answered

Marked out of 4.00

In the following belief network, S, L, C and F are binary random variables.



How many probabilities are needed to complete the belief network? What are they?

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A ▾

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x^2

Question 7

Not yet answered

Marked out of 1.00

If all the non-empty subsequences of a sequence are frequent in a sequence database, the sequence must be frequent in the database.

- ☐ True
- ☐ False

Question 8

Not yet answered

Marked out of 2.00

Assume that the range of an attribute is $[-200, 1800]$. If we would like to normalize this attribute so that its values fall into the range of $[0, 1]$ using the min-max normalization method, what value is value 200 converted to?

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A ▾

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Question 9

Not yet answered

Marked out of 1.00

Which of the following is NOT true about z-score?

- ☐ a. It measures how many standard deviations away a value is from the mean of a list of values.
- ☐ b. It can be used for normalizing attribute values.
- ☐ c. It can be used for detecting outliers.
- ☐ d. It is non-negative.
- ☐ e. It can only be used on numeric values.

Question 10

Not yet answered

Marked out of 1.00

The more accurate a neural network is on the training data set, the more accurate it will be on a test data set in the same domain.

Select one:

- ☐ True
- ☐ False

Question 11

Not yet answered

Marked out of 3.00

Each item in a store has a price, which is a positive value. Suppose that we are interested in finding all the frequent itemsets that not only satisfy a support threshold but also consist of items the sum of whose prices is less than \$500. How would you incorporate such a constraint into the FP-growth algorithm to find such itemsets efficiently? Please note that a solution that generates all the frequent itemsets and removes the ones the sum of whose prices is no less than \$500 is not considered an efficient solution. Also, we assume that the price list of items is stored in a table in memory, which can be accessed easily.

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Question 12

Not yet answered

Marked out of 1.00

What is the major advantage of recurrent neural networks compared to feed-forward neural networks?

Question 13

Not yet answered

Marked out of 1.00

What is the main mechanism used in transformers?

- ☐ a. Grounding Mechanism
- ☐ b. Attention Mechanism
- ☐ c. Reflection Mechanism
- ☐ d. Deflection Mechanism

Question 14

Not yet answered

Marked out of 1.00

An attribute can appear multiple times in a branch from the root to a leaf node in a decision tree.

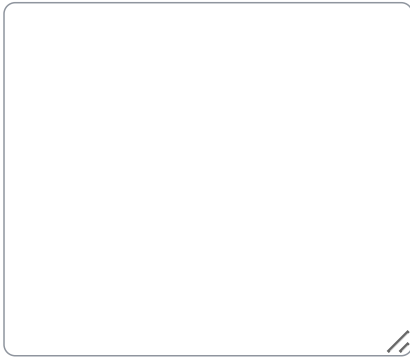
- ☐ True
- ☐ False

Question 15

Not yet answered

Marked out of 2.00

What will be the dimensions of the output feature map after applying 48 5x5 kernels to an input layer of 224x224 RGB images, with a stride of 4, assuming there is no padding?

**Question 16**

Not yet answered

Marked out of 1.00

Which of the following is **not true** about diffusion models?

- ☐ a. Diffusion models have large memory usage.
- ☐ b. Diffusion models are more likely to have mode collapse than GANs.
- ☐ c. Diffusion models are slow in inference.
- ☐ d. Diffusion models have a high computational cost.
- ☐ e. Diffusion models generate higher-quality images than earlier generative models such as GANs.

Question 17

Not yet answered

Marked out of 1.00

What is the purpose of using a hash-tree in the Apriori algorithm?

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Question 18

Not yet answered

Marked out of 1.00

Which of the following is NOT true for decision tree learning?

Select one:

- ☐ a. Decision tree learning can be used as a data preprocessing method for some other learning methods.
- ☐ b. Most decision tree learning algorithms employ a heuristic-based approach to guide its search in the hypothesis space.
- ☐ c. The major purpose of tree pruning is to make a decision tree simpler so that it is easier to understand.
- ☐ d. Most decision tree learning algorithms employ a top-down, recursive partitioning approach.
- ☐ e. It does not require any prior assumptions regarding the type of probability distributions satisfied by the class and other attributes.

Question 19

Not yet answered

Marked out of 2.00

What is the major cost of PrefixSpan?
What method did the authors of
PrefixSpan propose to reduce its major
cost?

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A ▾

B

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Question 20

Not yet answered

Marked out of 1.00

Which of the following is not true about node2vec?

Select one:

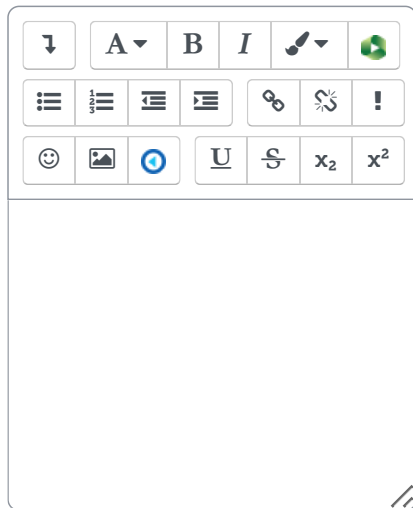
- ☐ a. node2vec performs well in both node classification and link prediction tasks.
- ☐ b. node2vec employs a second order random walk, where the next move depends on both the current and previous node.
- ☐ c. node2vec is limited in capturing local graph structures due to primarily focusing on higher-order structures by generating long walks.
- ☐ d. node2vec uses parameters, p and q , that allow control of both breadth-first and depth-first exploration.
- ☐ e. node2vec uses skip-gram with negative sampling.

Question 21

Not yet answered

Marked out of 1.00

What does BERT stand for and what benefit does it have over other transformer models?

A rich text editor interface. The toolbar at the top contains icons for undo, font color, bold, italic, link, unlink, and a green circular icon. Below the toolbar are three rows of icons: the first row has bulleted list, numbered list, decrease indent, increase indent, link, unlink, and a warning icon; the second row has an emoji icon, image upload, a blue circular icon, underline, strikethrough, subscript, and superscript. Below the toolbar is a large, empty text area for writing the answer.**Question 22**

Not yet answered

Marked out of 1.00

A knowledge graph-based recommender system can outperform a traditional recommender system, even on a sparse dataset.

Select one:

- ☐ True
- ☐ False

Question 23

Not yet answered

Marked out of 3.00

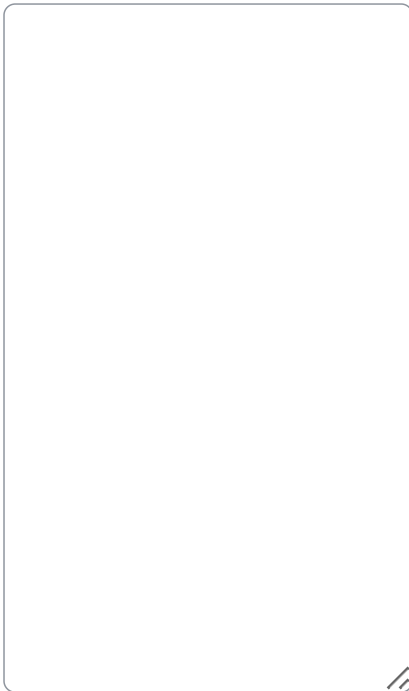
Given the following set of frequent 3-itemsets:

$\{a, b, c\}$, $\{a, b, d\}$, $\{a, b, e\}$, $\{a, c, d\}$,
 $\{a, c, e\}$,

$\{b, c, d\}$, $\{b, c, e\}$, $\{c, d, e\}$

Assume that all the length-3 frequent itemsets in a dataset are listed above.

What are the length-4 candidate itemsets generated by the Apriori algorithm? Show the major steps.



Question 24

Not yet answered

Marked out of 1.00

SBERT leverages a Siamese network architecture to compute fixed-size sentence embeddings by fine-tuning a pre-trained BERT model, enabling efficient pairwise semantic similarity comparisons.

Select one:

- ☐ True
- ☐ False

Question 25

Not yet answered

Marked out of 1.00

Deep learning algorithms learn deep decision trees.

Select one:

- ☐ True
- ☐ False

Question 26

Not yet answered

Marked out of 1.00

The minimum price of the items in an itemset is less than \$20 is a monotonic constraint.

- ☐ True
- ☐ False